



BITS Pilani

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CE F231 Fluid Mechanics

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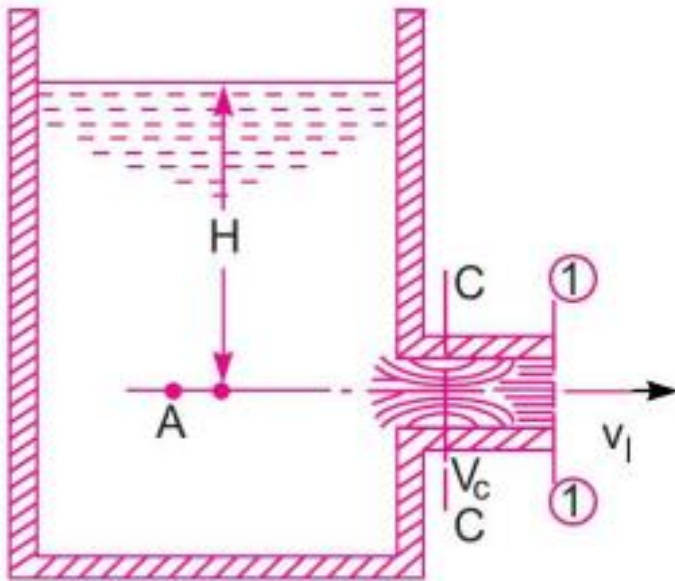
November 18, 2022



Lecture 27: Orifices and Mouthpieces

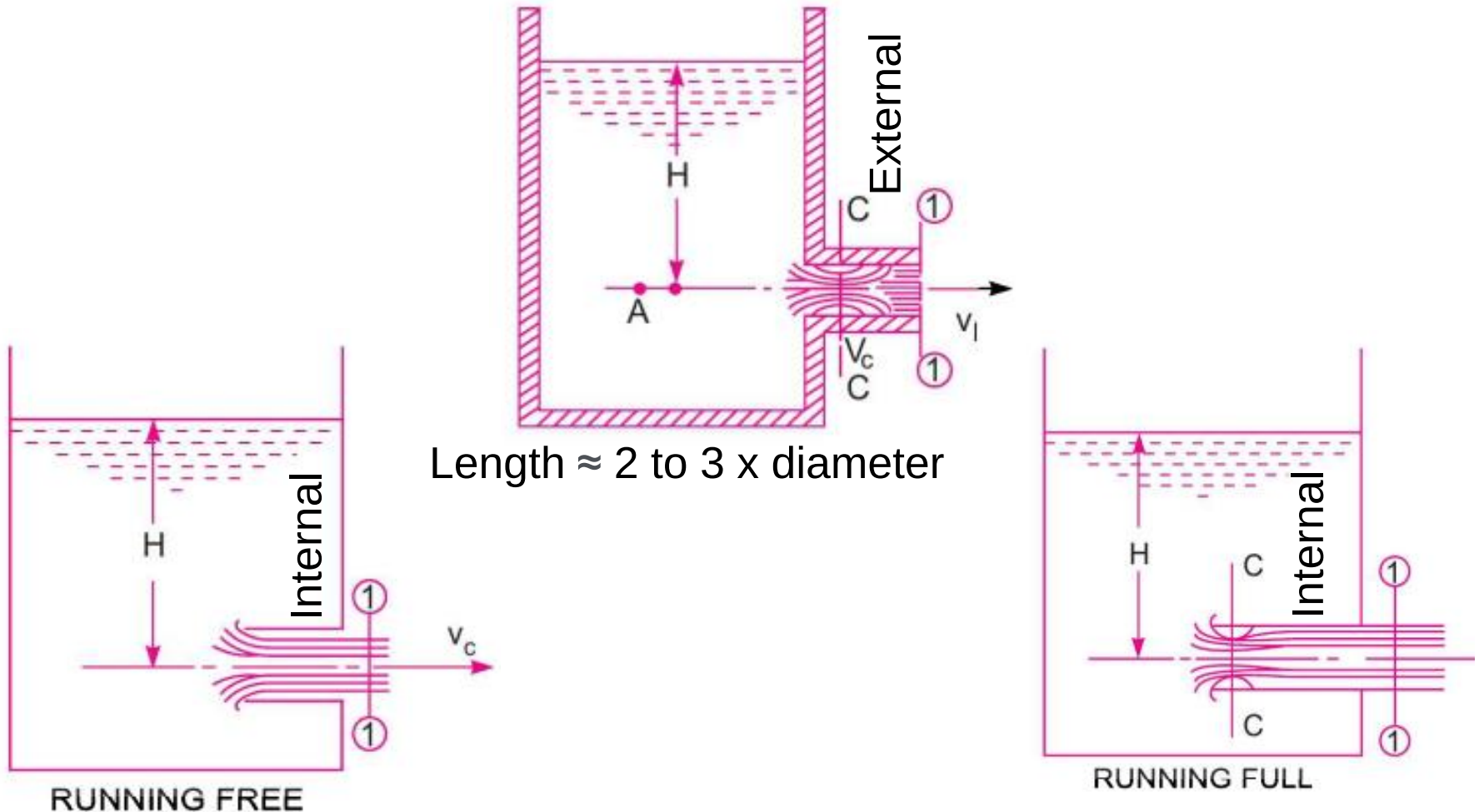
Mouthpieces

- Short length tube fitted to a circular orifice attached to a tank



Mouthpieces: Classification

- External and internal or re-entrant or Borda's mouthpiece

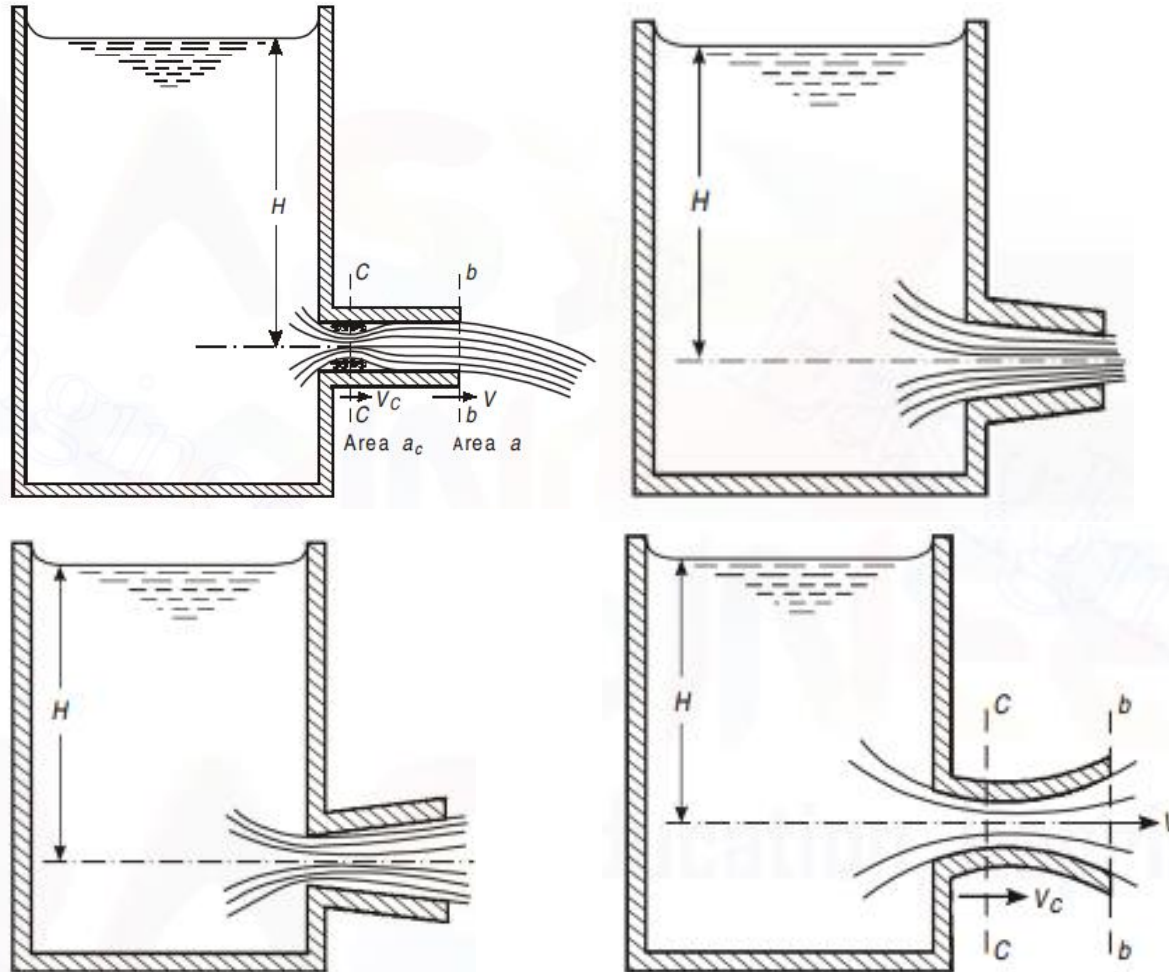


Length ≈ 1 x diameter

Length ≈ 3 x diameter

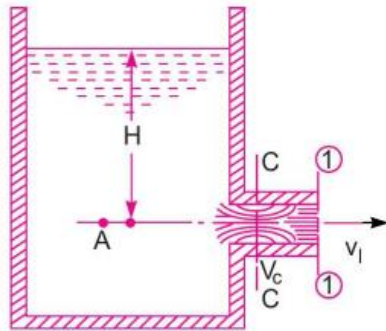
Mouthpieces: Classification

- Cylindrical, convergent, divergent, and convergent-divergent



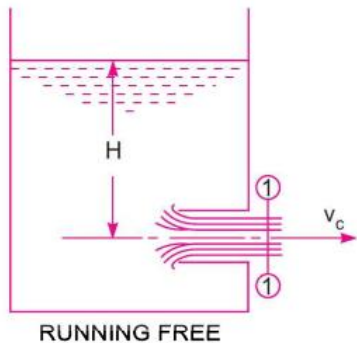
Mouthpieces: Discharge

External



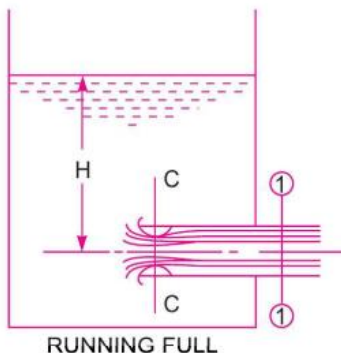
$$Q = 0.855 \times \text{Area of Mouthpiece} \times \sqrt{(2 g H)}$$

Internal



$$Q = 0.5 \times \text{Area of Mouthpiece} \times \sqrt{(2 g H)}$$

Internal



$$Q = 0.707 \times \text{Area of Mouthpiece} \times \sqrt{(2 g H)}$$

Mouthpieces: Discharge

Cylindrical external mouthpiece

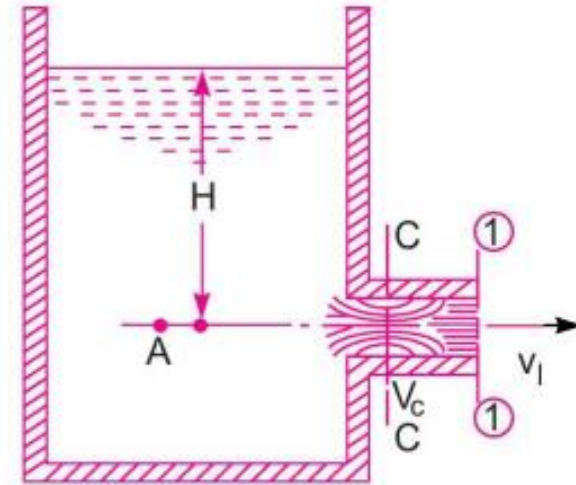
Applying Bernoulli's equation to point A and (1)-(1)

$$\frac{p_A}{\rho g} + \frac{v_A^2}{2g} + z_A = \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + h_L$$

$$h_L = \frac{(v_c - v_1)^2}{2g}$$

$$a_c \times v_c = a_1 v_1 \quad \Rightarrow \quad v_c = \frac{a_1 v_1}{a_c} = \frac{v_1}{a_c / a_1} \quad \Rightarrow \quad v_c = \frac{v_1}{0.62}$$

$$h_L = \frac{\left(\frac{v_1}{0.62} - v_1\right)^2}{2g} = \frac{v_1^2}{2g} \left[\frac{1}{0.62} - 1\right]^2 = \frac{0.375 v_1^2}{2g}$$



Mouthpieces: Discharge

Cylindrical external mouthpiece

Applying Bernoulli's equation to point A and (1)-(1)

$$\frac{p_A}{\rho g} + \frac{v_A^2}{2g} + z_A = \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + h_L$$

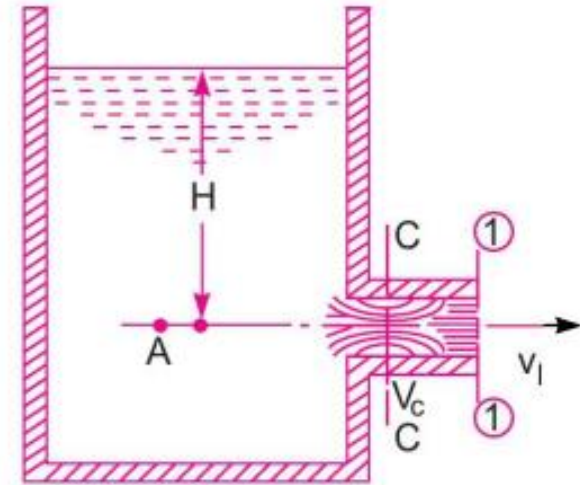
$$H + 0 = 0 + \frac{v_1^2}{2g} + .375 \frac{v_1^2}{2g}$$

$$H = 1.375 \frac{v_1^2}{2g}$$

$$v_1 = \sqrt{\frac{2gH}{1.375}} = 0.855 \sqrt{2gH}$$

$$C_v = 0.855; C_c = 1; C_d = 0.855$$

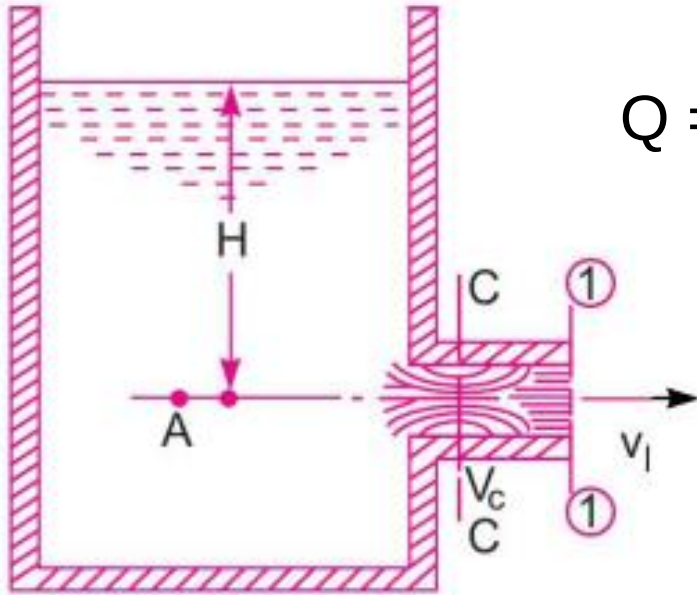
$$Q = 0.855 \times \text{Area of Mouthpiece} \times \sqrt{2gH}$$



$C_c = 1$ for jet emerging from external mouthpiece as cross sectional area of jet emerging from mouthpiece = cross sectional area of mouthpiece

Mouthpieces: Discharge

- Why discharge for external mouthpiece is greater than that obtained for orifice having same diameter?



$$Q = 0.855 \times \text{Area of Mouthpiece} \times \sqrt{2 g H}$$

Mouthpieces: Discharge

Borda's mouthpiece running free

Applying Bernoulli's equation to free surface of liquid and section (1)-(1)

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1$$

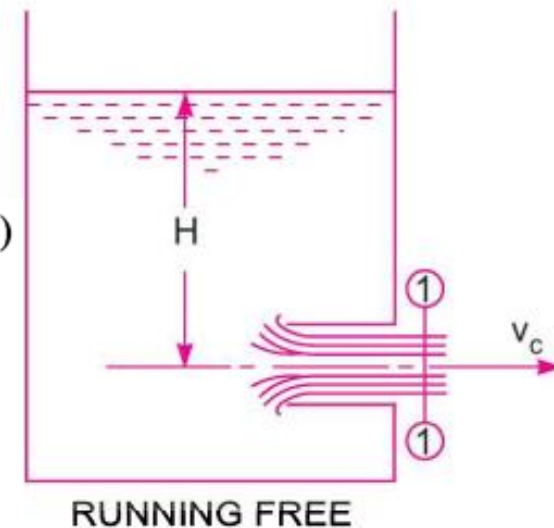
Taking the centre line of mouthpiece as datum, we have

$$z = H, z_1 = 0, \frac{p}{\rho g} = \frac{p_1}{\rho g} = p_{atmosp.} = 0,$$

$$v_1 = v_c, \quad v = 0$$

$$\therefore 0 + 0 + H = 0 + \frac{v_c^2}{2g} + 0 \quad \text{or} \quad H = \frac{v_c^2}{2g}$$

$$\therefore v_c = \sqrt{2gH}$$



No sudden expansion, no contact with mouthpiece wall, no loss $\Rightarrow C_v = 1$ 10

Mouthpieces: Discharge

Borda's mouthpiece running free

For calculating C_c

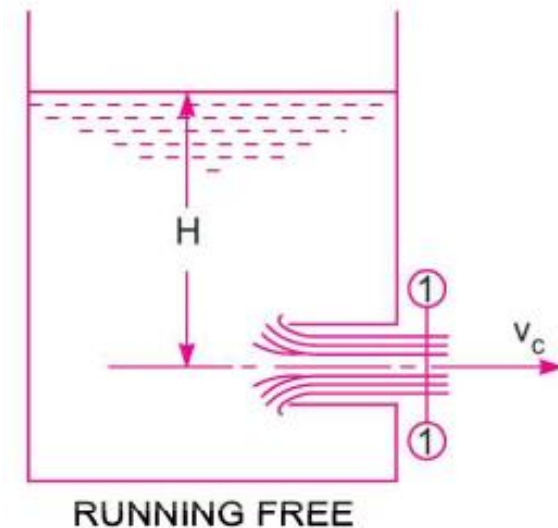
Use net force = rate of change in momentum

$$\rho g \cdot a \cdot H = \rho a_c \times v_c [v_c - 0]$$

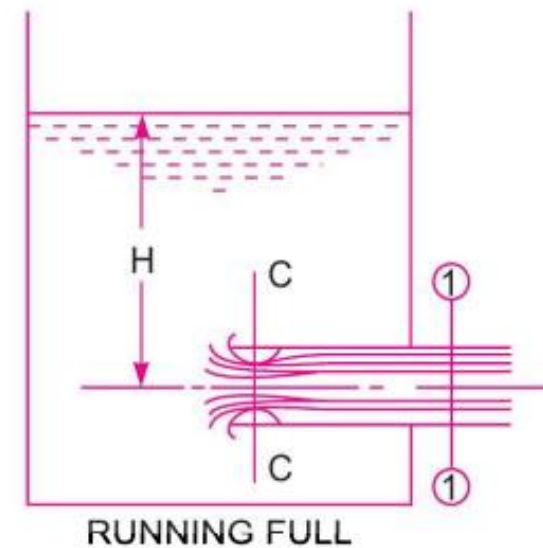
$$\rho g \cdot a \cdot H = \rho \cdot a_c \cdot 2g \cdot H$$

$$a = 2a_c \text{ or } \frac{a_c}{a} = \frac{1}{2} = 0.5 \Rightarrow C_c = \frac{a_c}{a} = 0.5$$

$$Q = 0.5 \times \text{Area of Mouthpiece} \times \sqrt{2gH}$$



Mouthpieces: Discharge



Borda's mouthpiece running full

Applying Bernoulli's equation to free surface of water in tank and section (1)-(1),

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + h_L$$

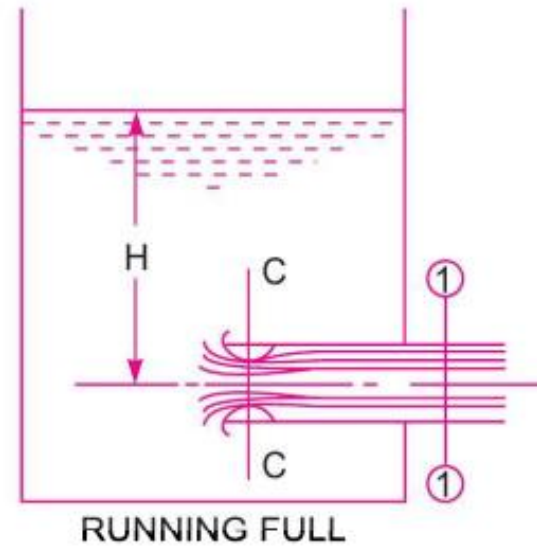
$$h_L = \frac{(v_c - v_1)^2}{2g}$$

$$a_c \times v_c = a_1 \times v_1 \Rightarrow v_c = \frac{a_1}{a_c} \times v_1 = \frac{v_1}{a_c / a_1} = \frac{v_1}{C_c} = \frac{v_1}{0.5} \Rightarrow v_c = 2v_1$$

the entrance to the mouthpiece then $a_c = C_c a$. At the entrance section of the mouthpiece since the flow pattern is same as that for the running free condition and hence

$$C_c = 0.5$$

Mouthpieces: Discharge



Borda's mouthpiece running full

Applying Bernoulli's equation to free surface of water in tank and section (1)-(1),

$$\frac{p}{\rho g} + \frac{v^2}{2g} + z = \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 + h_L$$

$$h_L = \frac{(2v_1 - v_1)^2}{2g} = \frac{v_1^2}{2g}$$

$$0 + 0 + H = 0 + \frac{v_1^2}{2g} + 0 + \frac{v_1^2}{2g}$$

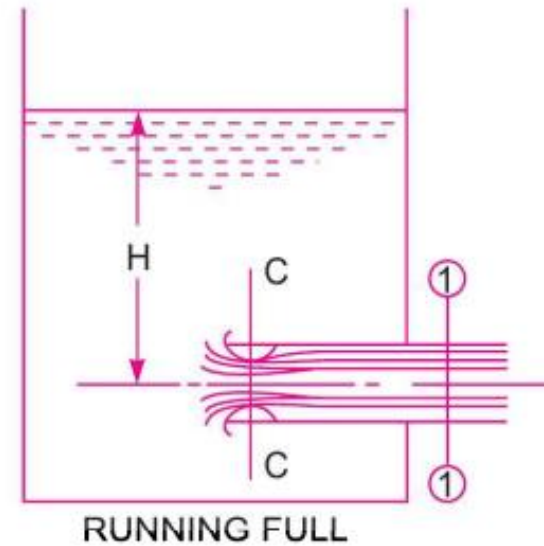
$$H = \frac{v_1^2}{2g} + \frac{v_1^2}{2g} = \frac{v_1^2}{g}$$

$$v_1 = \sqrt{gH}$$

$$C_v = \frac{v_1}{v_{th}} = \frac{\sqrt{gH}}{\sqrt{2gH}} = \frac{1}{\sqrt{2}} = 0.707$$

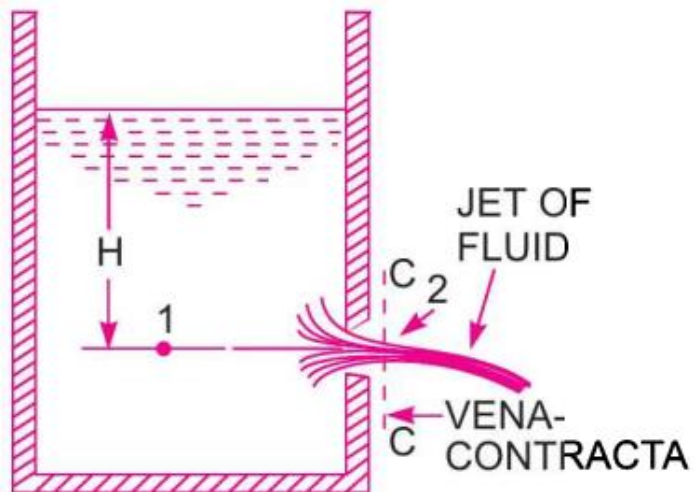
Mouthpieces: Discharge

Borda's mouthpiece running full

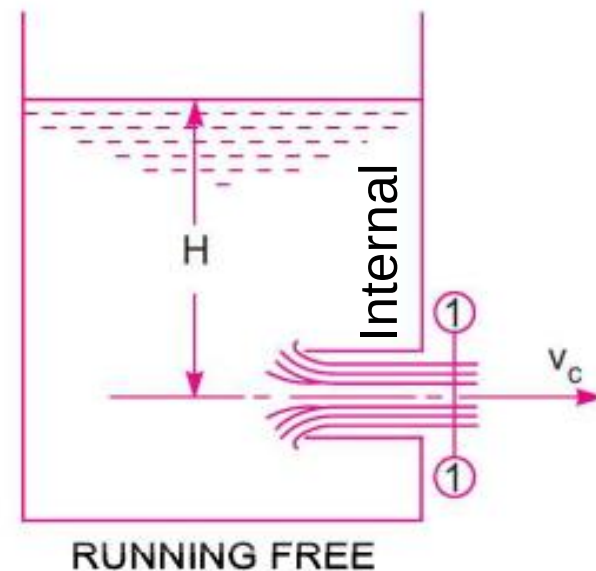


$$C_v = 0.707; C_c = 1; C_d = 0.707$$

$$Q = 0.707 \times \text{Area of Mouthpiece} \times \sqrt{2 g H}$$

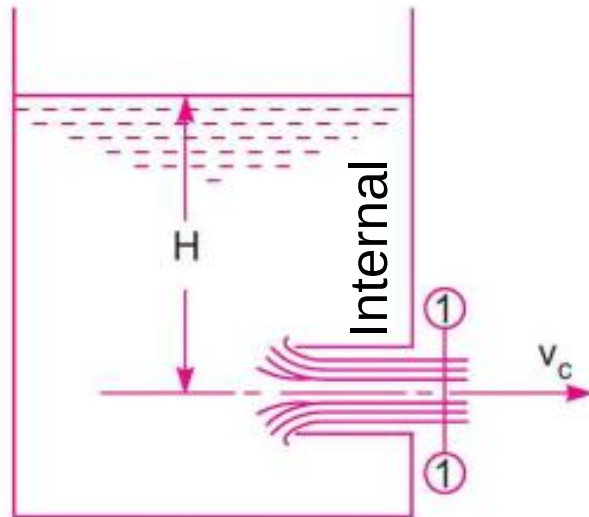


VS



RUNNING FREE

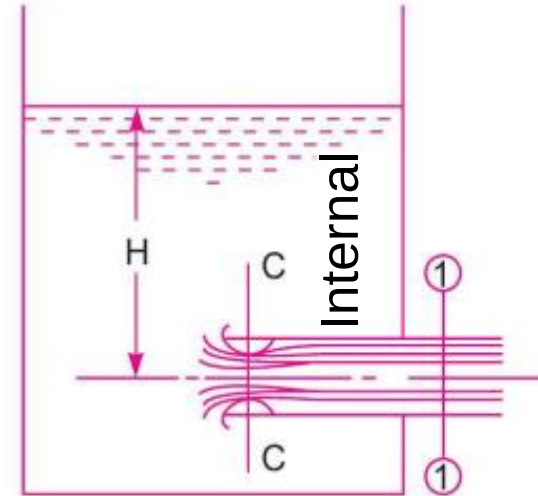
Length $\approx 1 \times$ diameter



RUNNING FREE

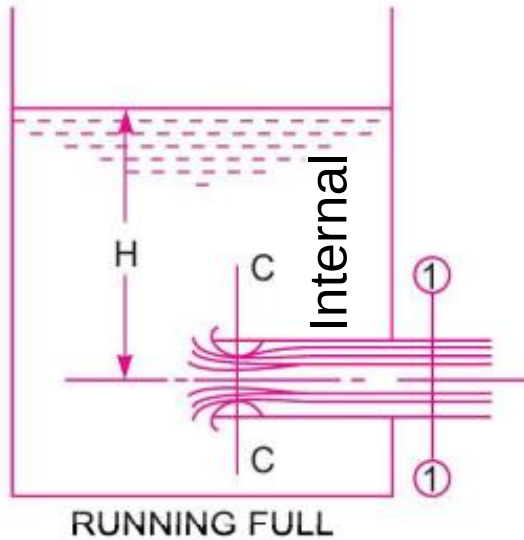
Length $\approx 1 \times$ diameter

VS



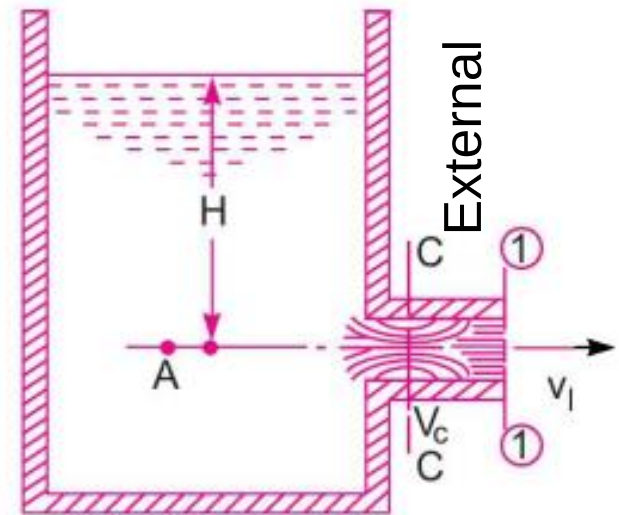
RUNNING FULL

Length $\approx 3 \times$ diameter



Length $\approx 3 \times$ diameter

VS



Length $\approx 2 \text{ to } 3 \times$ diameter



THANK YOU